B.2 Background on Input-to-state Stability & Notations

Background on Input-to-state Stability (ISS) In our proofs, the input-to-state stability (ISS) of a system can be established either through the definition of an ISS system in (Jiang et al., 1999, Def. 2.1) or via the use of an ISS-Lyapunov function as in (Jiang et al., 1999, Def. 2.2, Prop. 2.3). We also rely on the definition of a Lyapunov function and the difference Lyapunov equation for linear discrete-time homogeneous dynamical systems in (Gajic & Qureshi, 2008, Ch. 1, p. 8). The existence of a solution to the Lyapunov equation, together with its bound, is stated in (Gajic & Qureshi, 2008, Ch. 4, p. 110).

For reference, we briefly note that input-to-state stability (ISS) extends the classical notion of Lyapunov by explicitly accounting for external inputs: the state remains bounded and eventually whenever the input is bounded. A Lyapunov function provides an energy-like certificate for stability, while the associated Lyapunov equation offers a constructive method for obtaining such functions in linear settings. These notions are central for analyzing stability and will be used throughout our proofs.

Conventions and assumptions (used throughout). Let $\|\cdot\|$ denote the Euclidean norm on $\mathbb{R}^d$; for a matrix $M \in \mathbb{R}^{d\times d}$ we also write $\|M\|$ for the operator norm induced by the Euclidean norm, i.e. $\|M\| := \sup_{\|x\|=1}\|Mx\|$ (the spectral norm) (Horn & Johnson, 2012, pp. 343–346). We assume (i) $\sup_k \|\bar{\boldsymbol{A}}(k)\| < \infty$; (ii) $\boldsymbol{w}(k)$ is bounded (for a signal $\boldsymbol{w}$ we set $\|\boldsymbol{w}\|_\infty := \sup_{k\geq 0}\|\boldsymbol{w}(k)\|$) ; (iii) the controller gains are static and time-invariant scalar multiples of the identity. We use the standard meaning of the classes $\mathcal{K}$ and $\mathcal{KL}$ as in (Jiang et al., 1999).

B.3 Dynamics of the Average Error Across Layers

To formalize the problem setup, we consider N pairs of contrastive prompt/input tokens $\{(\boldsymbol{q}_i^+, \boldsymbol{q}_i^-)\}_{i=1}^N$, where $\boldsymbol{q}_i^+$ carries the desired property and $\boldsymbol{q}_i^-$ represents the opposite. For discrete time (layer) k, let $\boldsymbol{x}_i^\pm(k) \in \mathbb{R}^d$ denote the corresponding activation vectors. The layer-to-layer evolution is

$$\boldsymbol{x}_i(k+1) = f_i^{(k)}\big(\boldsymbol{x}(k)\big), i=1,\ldots,N, \tag{24}$$

with $f_i^{(k)} : \mathbb{R}^d \to \mathbb{R}^d$ differentiable on the operating region. A steering input $u(t)$ is applied on the undesired branch:

$$\boldsymbol{x}_i^-(k+1) = f_i^{(k)}\big(\boldsymbol{x}_i^-(k) + \boldsymbol{u}(k)\big). \tag{25}$$

Defining $\bar{\boldsymbol{x}}^\pm(k) := \frac{1}{N}\sum_{i=1}^N \boldsymbol{x}_i^\pm(k)$, we track the per-pair and average errors as

$$\boldsymbol{e}_i(k) := \boldsymbol{x}_i^+(k) - \boldsymbol{x}_i^-(k), \qquad \bar{\boldsymbol{e}}(k) := \bar{\boldsymbol{x}}^+(k) - \bar{\boldsymbol{x}}^-(k), \qquad \tilde{\boldsymbol{e}}_i(k) = \boldsymbol{e}_i(k) - \bar{\boldsymbol{e}}(k). \tag{26}$$

Furthermore, we define $\boldsymbol{A}_i(k)$ as the Jacobian of $f_i^{(k)}$ at $\boldsymbol{x}_i^+(k)$:

$$\boldsymbol{A}_i(k) := J_{f_i^{(k)}}\big(\boldsymbol{x}_i^+(k)\big), \qquad \bar{\boldsymbol{A}}(t) := \frac{1}{N}\sum_{i=1}^N \boldsymbol{A}_i(k), \qquad \tilde{\boldsymbol{A}}_i(k) = \boldsymbol{A}_i(k) - \bar{\boldsymbol{A}}(k). \tag{27}$$

The dynamic of the average error $\bar{\boldsymbol{e}}(k)$ is then given by Proposition 2.

Proposition 2 (Error dynamics of activation steering) *The error dynamics $\bar{\boldsymbol{e}}(k)$ in activation steering is of the form:*

$$\bar{\boldsymbol{e}}(k+1) = \bar{\boldsymbol{A}}(k)\bar{\boldsymbol{e}}(k) - \bar{\boldsymbol{A}}(k)\boldsymbol{u}(k) + \boldsymbol{w}(k), \tag{20}$$

where $\bar{\boldsymbol{A}}(k)$ is the mean local Jacobian of $f_i^{(k)}$ at $\boldsymbol{x}_i^+(k)$ and the disturbance term $\boldsymbol{w}(k)$ collects heterogeneity. See Appendix B.3 for detailed proof and explanations of the terms.

Proof. The evolution of the average error $\bar{\boldsymbol{e}}(k)$ through layers can be described as follows:

$$\bar{\boldsymbol{e}}(k+1) = \bar{\boldsymbol{x}}^+(k+1) - \bar{\boldsymbol{x}}^-(k+1) = \frac{1}{N}\sum_{i=1}^N \Big[f_i^{(k)}\big(\boldsymbol{x}_i^+(k)\big) - f_i^{(k)}\big(\boldsymbol{x}_i^-(k) + \boldsymbol{u}(k)\big)\Big]. \tag{28}$$

Linearizing $f_i^{(k)}$ around $\boldsymbol{x}_i^+(k)$, we obtain

$$f_i^{(k)}\big(\boldsymbol{x}_i^+(k) + \delta\big) \approx f_i^{(k)}\big(\boldsymbol{x}_i^+(k)\big) + J_{f_i^{(k)}}\big(\boldsymbol{x}_i^+(k)\big)\cdot\delta, \tag{29}$$

where $J_{f_i^{(k)}}$ denotes the Jacobian of $f_i^{(k)}$.

Setting $\delta = -\boldsymbol{e}_i(k) + \boldsymbol{u}(k)$ yields

$$f_i^{(k)}\big(\boldsymbol{x}_i^+(k)+\delta\big) \approx f_i^{(k)}\big(\boldsymbol{x}_i^+(k)\big) + \boldsymbol{A}_i(k)\big(\boldsymbol{e}_i(k)+\boldsymbol{u}(k)\big),$$

Insert this into Eqn.28 we obtain

$$\bar{\boldsymbol{e}}(k+1) = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{A}_i(k)\boldsymbol{e}_i(k) - \bar{\boldsymbol{A}}(k)\boldsymbol{u}(k). \tag{30}$$

Recall that

$$\boldsymbol{e}_i(k) = \bar{\boldsymbol{e}}(k) + \tilde{\boldsymbol{e}}_{(i)}(k), \text{ then } \frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{e}}_{(i)}(k) = 0,$$

and

$$\boldsymbol{A}_{(i)}(t) = \bar{\boldsymbol{A}}(k) + \tilde{\boldsymbol{A}}_{(i)}(k), \text{ then } \frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{A}}_{(i)}(k) = 0.$$

Therefore,

$$\begin{aligned}\bar{e}(k+1) &= \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{A}_i(k)\boldsymbol{e}_i(k) - \bar{\boldsymbol{A}}(k)\boldsymbol{u}(k)\\ &= \frac{1}{N}\sum_{i=1}^{N}\bar{\boldsymbol{A}}(k)\bar{\boldsymbol{e}}_i(k) - \bar{\boldsymbol{A}}(k)\boldsymbol{u}(k) + \frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{e}}_{(i)}(k)\tilde{\boldsymbol{A}}_{(i)}(k)\\ &\underbrace{+\bar{\boldsymbol{A}}(k)\frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{e}}_{(i)}(k) + \bar{\boldsymbol{e}}(k)\frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{A}}_{(i)}(k)}_{=0}\end{aligned} \tag{31}$$

We then obtain the final state-space model for the dynamics of $\bar{e}(t)$ as

$$\bar{\boldsymbol{e}}(k+1) = \bar{\boldsymbol{A}}(k)\bar{e}(k) - \bar{\boldsymbol{A}}(k)\boldsymbol{u}(k) + \boldsymbol{w}(k), \tag{32}$$

where

$$\boldsymbol{w}(k) = \frac{1}{N}\sum_{i=1}^{N}\tilde{\boldsymbol{A}}_i(k)\tilde{\boldsymbol{e}}_i(k),$$

which acts as a time-dependent exogeneous disturbance to the model □

B.4 Proportional (P) Control

Consider proportional control with

$$\boldsymbol{u}(k) = K_P\bar{\boldsymbol{e}}(k), \quad (K_I = K_D = 0).$$

The dynamics Eqn. 20 then become

$$\bar{\boldsymbol{e}}(k) = M_P(k)\bar{\boldsymbol{e}}(k) + \boldsymbol{w}(k), \tag{33}$$

where $M_P(k) = \bar{\boldsymbol{A}}(k)(I - K_P)$.

With a suitable choice of K_P, the system can be made input-to-state stable (ISS); that is, there exist a $\mathcal{KL}$-function β and a $\mathcal{K}$-function γ such that, for all disturbance $\boldsymbol{w}$ with bounded sup norm and all initial states $\bar{\boldsymbol{e}}(0)$,

$$\|\bar{\boldsymbol{e}}(k)\| \le \beta(\|\bar{\boldsymbol{e}}(0)\|, k) + \gamma(\|\boldsymbol{w}\|_\infty), \quad k \in \mathbb{Z}_{\ge 0}, \tag{34}$$

see (Jiang et al., 1999, Def. 2.1).

In particular, the error decays from the initial condition and remains bounded under bounded disturbances.

Proposition 1 (Steady-state error of P-control activation steering) *P-control activation steering ensures input-to-state stability (ISS) for an appropriate range of K_p. However, there still exists a steady-state error due to the disturbance $\boldsymbol{w}(k)$ to the state of the system. In the best case, when $\boldsymbol{w}(k)$ converges to $\boldsymbol{w}$, under a mild condition, the expected error, i.e., the difference-in-means, $\boldsymbol{r}(k) = \bar{\boldsymbol{e}}(k)$ eventually converges to a steady state $\bar{\boldsymbol{e}}_{ss} \propto \boldsymbol{w}$. Therefore, $\bar{\boldsymbol{e}}_{ss} \neq 0$ if $\boldsymbol{w} \neq 0$.*

Proof. Assume $\sup_k \|\bar{\boldsymbol{A}}(k)\| \leq M < \infty$, $K_P = pI$ with $p > 0$. Since $M_P(k) = \bar{\boldsymbol{A}}(k)(I - K_P) = \bar{\boldsymbol{A}}(k)(1-p)I$, by sub-multiplicative property of matrix norm we have

$$\|M_P(t)\| \leq \|\bar{A}(t)\| \|(1-p)I\| \leq M|1-p| =: q. \tag{35}$$

For $p \in \left(1-\frac{1}{M}, 1+\frac{1}{M}\right)$, we have $q < 1$.

Expanding recursively,

$$\bar{\boldsymbol{e}}(k) = M_P(k-1)\cdots M_P(0)\bar{\boldsymbol{e}}(0) + \sum_{j=0}^{k-1} M_P(k-1)\cdots M_P(j+1)w(j).$$

Hence,

$$\|\bar{\boldsymbol{e}}(k)\| \leq q^k \|\bar{\boldsymbol{e}}(0)\| + \sum_{j=0}^{k-1} q^{k-1-j}\|w(j)\| \leq q^k \|\bar{e}(0)\| + \frac{1-q^k}{1-q}\|w\|_\infty \leq q^k \|\bar{e}(0)\| + \frac{1}{1-q}\|w\|_\infty. \tag{36}$$

Since $q < 1$, we can set $\beta(s,k) = q^k s$, which is a $\mathcal{KL}$-function (decaying to zero as $k \to \infty$), and $\gamma(s) = \frac{1}{1-q}s$, which is a $\mathcal{K}$-function, satisfying Eqn. 34. Therefore, the system is ISS.

However, there exists a steady-state error due to the disturbance $\boldsymbol{w}(k)$. In the best case, when $\bar{\boldsymbol{A}}(k)$ converges to $\bar{\boldsymbol{A}}$ and $\boldsymbol{w}(k)$ converges to $\boldsymbol{w}$, the error $\bar{\boldsymbol{e}}(k)$ eventually converges to a steady state given by

$$\bar{\boldsymbol{e}}_{ss} = (I - \bar{\boldsymbol{A}}(1-pI))^{-1}\boldsymbol{w}.$$

Therefore, $\bar{\boldsymbol{e}}_{ss} \neq 0$ if $\boldsymbol{w} \neq 0$. □

Remark 1 (Convergence rate versus K_P.) *From Ineq. 36, smaller q yields faster convergence. Because*

$$q(p) = M|1-p| = \begin{cases} M(1-p), & p \in \left(1-\frac{1}{M}, 1\right), \\ M(p-1), & p \in \left[1, 1+\frac{1}{M}\right), \end{cases}$$

we have $\frac{d}{dp}q(p) = -M < 0$ for $p < 1$ and $\frac{d}{dp}q(p) = M > 0$ for $p > 1$. Therefore the contraction factor $q(p)$ is minimized at

$$p^\star = 1 \quad \Longrightarrow \quad q^\star = 0,$$

and increases as p moves away from 1 within the admissible interval.

B.5 Proportional-Integral (PI) Control

To reduce the steady-state error, the proportional controller is extended with an integral action, resulting in a proportional-integral (PI) control law:

$$\boldsymbol{u}(k) = K_p\bar{\boldsymbol{e}}(k) + K_I\boldsymbol{s}(k), \qquad \boldsymbol{s}(k+1) = \boldsymbol{s}(k) + \bar{\boldsymbol{e}}(k), \qquad (K_D = 0). \tag{37}$$

The dynamics Eqn. 20 then become

$$\bar{\boldsymbol{e}}(k+1) = \bar{\boldsymbol{A}}(k)(I - K_P)\bar{\boldsymbol{e}}(k) - \bar{\boldsymbol{A}}(k)K_I\boldsymbol{s}(k) + \boldsymbol{w}(k). \tag{38}$$

We use the following orthogonal decomposition for $\boldsymbol{w}(k)$:

$$\boldsymbol{w}(k) = \boldsymbol{w}^{\parallel}(k) + \boldsymbol{w}^{\perp}(k),$$

where $\boldsymbol{w}^{\parallel}(k) \in \mathrm{Im}\bar{\boldsymbol{A}}(k)$ and $\boldsymbol{w}^{\perp}(k) \in (\mathrm{Im}\bar{\boldsymbol{A}}(k))^{\perp}$.

The impact of $\boldsymbol{w}^{\parallel}(k)$ on the error can be eliminated by PI control, as discussed below. On the other hand, P-only control is not able to do so, because keeping $\bar{\boldsymbol{e}}(k) = 0$ requires $\boldsymbol{u}(k) = 0$, leaving no component in $\boldsymbol{u}(k)$ that can compensate for $\boldsymbol{w}^{\parallel}(k)$.

Since $\boldsymbol{w}^{\parallel}(k) \in \mathrm{Im}\bar{\boldsymbol{A}}(k)$, it can be expressed as

$$\boldsymbol{w}^{\parallel}(k) = \bar{\boldsymbol{A}}(k)K_I s^*(k). \quad \Longleftrightarrow \quad \boldsymbol{s}^*(k) = K_I^{-1}\bar{\boldsymbol{A}}(k)^{\dagger}\boldsymbol{w}^{\parallel}(k)$$

Let $\tilde{\boldsymbol{s}}(k) = \boldsymbol{s}(k) - \boldsymbol{s}^*(k)$ and $d(k) = s^*(k+1) - s^*(k)$. Therefore,

$$\tilde{\boldsymbol{s}}(k+1) = \tilde{\boldsymbol{s}}(k) + \bar{\boldsymbol{e}}(k) - \boldsymbol{d}(k) \tag{39}$$